Statement of Purpose

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My interest in machine learning started with Kaggle competitions, where I began experimenting with different models and learning how to improve their performance. Initially, I focused mainly on getting better results. However, as I worked on more complex projects, I became interested in another question: how can we make a model not only accurate but also faster and less expensive to run? I realised that a model is not always useful simply because it performs well. It also needs to work within real-world limitations, especially when computing resources are limited. This is what gradually led me towards efficient model inference, which I now want to explore further through graduate research.

I am currently pursuing a B.Sc. in Electrical and Electronic Engineering at Khulna University of Engineering & Technology (KUET), Bangladesh. Alongside my studies, I have developed my skills in machine learning through personal projects, competitions, and research collaborations. One important experience was participating in BUET CSE Fest DL Sprint 4.0, where my team worked on Bengali speech recognition. We reduced the processing time for 22 hours of audio from approximately four hours to just 26 minutes. We also decided not to use an additional audio-processing step because the small improvement in accuracy did not justify the extra processing time. Our work received Second Runner-Up and the Dataset Prize. This experience helped me understand that improving a model is not just about achieving better accuracy; it is also about making it practical to use.

My interest in this area grew further when I worked on adapting a speech recognition model to Bengali. I found that the way the model divided Bengali words into smaller units made it difficult for the model to recognise speech efficiently. Instead of making the model larger, I changed how it represented Bengali words. This reduced the length of the sequences the model had to process and improved its decoding behaviour. Our work was presented as a poster at the MusIML Workshop at ICML 2026. It made me more curious about how small, carefully chosen changes can make AI models more efficient without requiring substantially more computing power.

I have also contributed to creating Lipi-Ghor, an open Bengali speech dataset containing approximately 882 hours of recordings. The associated data paper was accepted to the SALMA Workshop at EMNLP 2026. Currently, I lead a six-member university team in the NeurIPS Speech Accessibility Challenge, where we are working on speech recognition that can operate under CPU-only computing constraints. These

experiences have strengthened my interest in developing AI systems that are not only effective but also accessible to people who do not have expensive hardware.

At FIT, I hope to develop a deeper understanding of how AI models can be made faster, more efficient, and more reliable. I am particularly interested in reducing the computational cost of running large models while maintaining their performance. Although my current work focuses mainly on speech recognition, I would like to explore these challenges in other AI applications as well. In the long term, I hope to build a career in AI research, investigating practical ways to make advanced models more accessible and useful.

My journey from Kaggle competitions to research has taught me that there is more to machine learning than achieving a high score. I want to understand why models behave the way they do, identify their limitations, and find better ways to make them work in practice. Graduate study at FIT would be an important step towards developing the knowledge and research skills needed to pursue these goals.